A gcd-sum weighted by the character mod 4: a proof of the Yanev–Kotesovec conjecture on OEIS A327123, with a correction to the entry's multiplicative formula

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Abstract

OEIS A327123 is defined by the generating function $\sum_{k\geq 1}\varphi(k)x^k/(1+x^{2k})$. In June 2024 Velin Yanev and Vaclav Kotesovec conjectured on the entry that $a(n)=\sum_{k=1}^{n}\sin(\gcd(k,n)\pi/2)$. We prove it. The weight $\sin(d\pi/2)$ is the non-principal Dirichlet character χ modulo 4, and both sides turn out to be the Dirichlet convolution $\chi*\varphi$: the conjectured side by grouping the summation range by $\gcd(k,n)$, the defining side by expanding the generating function as a geometric series. We also record a correction: the multiplicative formula currently posted on the entry is false at every prime $p\equiv 1 \pmod 4$, first at $n=5$, where it gives 1 against the entry's own value $a(5)=5$. The correct local value is p^e, and we derive all three local values from the convolution.

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1 The sequence and the conjecture

Throughout p is a prime and φ is Euler's totient function (OEIS A000010). Let

$$\chi(d)=\sin\left(\frac{d\pi}{2}\right)=\begin{cases}0, & d\equiv 0 \pmod 2,\\ 1, & d\equiv 1 \pmod 4,\\ -1, & d\equiv 3 \pmod 4,\end{cases}$$

which is the non-principal Dirichlet character modulo 4. In particular χ is completely multiplicative and takes only the values $0,\pm 1$, so every expression below is an identity between integers despite the trigonometric notation.

OEIS A327123 is defined by

$$\sum_{n\geq 1} a(n)\,x^n \;=\; \sum_{k\geq 1}\frac{\varphi(k)\,x^k}{1+x^{2k}}, \tag{1}$$

with offset 1, and begins

$$a(1),\ldots,a(12) \;=\; 1,\,1,\,1,\,2,\,5,\,1,\,5,\,4,\,5,\,5,\,9,\,2.$$

The entry carries the following line in its FORMULA section:

> **Conjecture:** $a(n)=\mathrm{Sum}_{k=1..n}\ \sin(\mathrm{GCD}(k,n){*}\mathrm{Pi}/2)$. — *Velin Yanev* and *Vaclav Kotesovec*, Jun 01 2024

As of the "Last modified" line on the live entry (1 June 2024, revision 22) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs. Write

$$S(n) = \sum_{k=1}^{n} \chi\bigl(\gcd(k,n)\bigr). \tag{2}$$

The claim is that $S(n) = a(n)$ for all $n \geq 1$.

2 Two lemmas

Lemma 1. *Let f be any arithmetic function and $n \geq 1$. Then $\sum_{k=1}^{n} f(\gcd(k,n)) = \sum_{d|n} f(d)\varphi(n/d) = (f * \varphi)(n)$, where $*$ is Dirichlet convolution.*

Proof. Each $k \in \{1,\ldots,n\}$ satisfies $\gcd(k,n) = d$ for exactly one divisor $d \mid n$, so the range splits as a disjoint union and $\sum_{k=1}^{n} f(\gcd(k,n)) = \sum_{d|n} f(d)\,\#\{k \leq n : \gcd(k,n) = d\}$. Writing $k = dm$, the conditions $1 \leq k \leq n$ and $\gcd(k,n) = d$ become $1 \leq m \leq n/d$ and $\gcd(m, n/d) = 1$, so the inner count is $\varphi(n/d)$. As d runs over the divisors of n, so does n/d. □

Lemma 2. *The sequence defined by (1) satisfies $a = \chi * \varphi$, that is, $a(n) = \sum_{d|n} \chi(d)\varphi(n/d)$ for every $n \geq 1$.*

Proof. Expand each summand of (1) as a geometric series in x^{2k}:

$$\frac{\varphi(k)x^k}{1+x^{2k}} = \varphi(k)\,x^k \sum_{j\geq 0} (-1)^j x^{2kj} = \sum_{j\geq 0} (-1)^j \varphi(k)\,x^{k(2j+1)}.$$

Summing over k, the coefficient of x^n receives a contribution $(-1)^j\varphi(k)$ from each pair (k,j) with $k(2j+1) = n$. Such pairs correspond exactly to the factorisations $n = k\,m$ with $m = 2j+1$ odd, and then $j = (m-1)/2$. Hence

$$a(n) = \sum_{\substack{m|n \\ m \text{ odd}}} (-1)^{(m-1)/2}\,\varphi(n/m).$$

For odd m we have $(-1)^{(m-1)/2} = 1$ when $m \equiv 1 \pmod 4$ and -1 when $m \equiv 3 \pmod 4$, which is $\chi(m)$; and $\chi(m) = 0$ for even m, so the even divisors may be admitted to the sum without changing it. Therefore $a(n) = \sum_{m|n} \chi(m)\varphi(n/m)$. □

3 The theorem

Theorem 3. *For every $n \geq 1$,*

$$\sum_{k=1}^{n} \sin\left(\frac{\gcd(k,n)\pi}{2}\right) = a(n),$$

where a is defined by (1). That is, the Yanev–Kotesovec conjecture on OEIS A327123 is true.

Proof. Lemma 1 applied to $f = \chi$ gives $S = \chi * \varphi$, and Lemma 2 gives $a = \chi * \varphi$. Hence $S = a$. □

The two lemmas are independent of each other: one is a statement about counting k by $\gcd(k,n)$, the other about expanding a generating function. The conjecture is exactly the assertion that they meet, and the common value $\chi * \varphi$ is what they meet at.

4 The local values, and a correction to the entry

Since χ and φ are multiplicative, so is $a = \chi * \varphi$; the entry already carries the keyword `mult`. We compute the local values, because the ones currently posted on the entry are not all correct.

Corollary 4. *For every $e \geq 1$,*

$$a(2^e) = 2^{e-1}, \qquad a(p^e) = p^e \quad (p \equiv 1 \bmod 4), \qquad a(p^e) = \frac{p^{e+1} - p^e + 2(-1)^e}{p+1} \quad (p \equiv 3 \bmod 4).$$

Proof. By Lemma 2, $a(p^e) = \sum_{i=0}^{e} \chi(p^i)\varphi(p^{e-i})$.

If $p = 2$ then $\chi(2^i) = 0$ for $i \geq 1$, so only $i = 0$ survives and $a(2^e) = \varphi(2^e) = 2^{e-1}$.

If $p \equiv 1 \pmod 4$ then $p^i \equiv 1 \pmod 4$ for all i, so $\chi(p^i) = 1$ and $a(p^e) = \sum_{i=0}^{e} \varphi(p^{e-i}) = \sum_{d|p^e} \varphi(d) = p^e$, by the classical identity $\sum_{d|m} \varphi(d) = m$.

If $p \equiv 3 \pmod 4$ then $p^i \equiv 3 \pmod 4$ for odd i and $\equiv 1$ for even i, so $\chi(p^i) = (-1)^i$. Substituting $m = e - i$,

$$a(p^e) = \sum_{i=0}^{e} (-1)^i \varphi(p^{e-i}) = (-1)^e \Big[1 + \sum_{m=1}^{e} (-1)^m \big(p^m - p^{m-1}\big)\Big].$$

The inner sum is $\frac{p-1}{p} \sum_{m=1}^{e} (-1)^m p^m = \frac{p-1}{p} \cdot \frac{-p\big(1-(-p)^e\big)}{1+p} = -\frac{(p-1)\big(1-(-p)^e\big)}{p+1}$, so

$$a(p^e) = (-1)^e \cdot \frac{(p+1) - (p-1) + (p-1)(-p)^e}{p+1} = \frac{2(-1)^e + (p-1)\,p^e}{p+1},$$

using $(-1)^e(-p)^e = p^e$. This is the stated expression. □

Remark 5 (Correction)**.** The FORMULA section of A327123 currently states, in a contribution of Amiram Eldar dated 28 August 2023:

> Multiplicative with $a(2^e) = 2^{(e-1)}$, and if p is an odd prime $a(p^e) = 1$ if $p \equiv 1 \pmod 4$ and $(p^{(e+1)} - p^e + 2(-1)^e)/(p+1)$ otherwise.

The branch for $p \equiv 1 \pmod 4$ is false. By Corollary 4 the correct value there is p^e, not 1. The discrepancy is visible in the entry's own DATA section at the very first odd prime $p \equiv 1 \pmod 4$: the posted formula gives $a(5) = 1$, whereas the entry lists $a(5) = 5$. The literal formula disagrees with the published data at $n = 5, 10, 13, 15, 17, 20, 25, 26, \ldots$ — that is, at every n divisible by a prime $p \equiv 1 \pmod 4$. The $p = 2$ and $p \equiv 3 \pmod 4$ branches are correct as posted, and are confirmed by Corollary 4. The error appears to be a transcription slip for p^e; it does not affect the conjecture, which concerns a different formula.

5 Verification

Every claim above was checked numerically, by routines written separately from this note and by algorithms that share no code path.

The conjectured gcd-sum (2) was evaluated directly, forming $\gcd(k, n)$ for $k = 1, \ldots, n$ and applying χ as an integer-valued function of the residue mod 4. The defining sequence was computed independently by expanding (1) as a power series, accumulating $(-1)^j \varphi(k)$ into the coefficient of $x^{k(2j+1)}$; this route uses the generating function only and never forms a gcd. The two agree with each other and with all 69 terms of the entry's DATA section, over its full range $n = 1, \ldots, 69$; the gcd-sum and the convolution $\sum_{d|n} \chi(d)\varphi(n/d)$ agree for all $n \leq 800$.

The local values of Corollary 4 were checked against the data over the same range and agree throughout. The posted formula of Remark 5, implemented exactly as written, fails against the entry's DATA first at $n = 5$ (formula 1, data 5) and then at $n = 10, 13, 15, 17, 20, 25, 26, \ldots$, as stated.

6 Concluding remarks

The conjecture is elementary once the character is recognised: writing $\sin(d\pi/2) = \chi(d)$ turns a trigonometric-looking statement into a Dirichlet convolution, and both sides then reduce to $\chi * \varphi$ by one lemma each. What makes it worth recording is that the entry has carried the statement as a conjecture since June 2024 while also carrying, since September 2019, the equivalent description "Moebius transform of A050469".

The same gcd-sum lemma settles the conjectures on OEIS A358272 and A358319, both due to Werner Schulte, where the weights are $d\lambda(d)$ and $d\,\mu(\mathrm{rad}(d))$ instead of $\chi(d)$; those are treated separately. In each case the entry's own FORMULA section already contains the convolution, and the conjecture is the gcd-sum form of it.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A327123, definition of I. Gutkovskiy, 14 September 2019, formula of A. Eldar, 28 August 2023, and conjecture of V. Yanev and V. Kotesovec, 1 June 2024; sequences A000010, A050469, A358272, A358319.